Why Does $d^2 = 0$ Encode Higher Information?

Morse Continuation Maps, Twisted Complexes, and Higher Homotopic Data

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This is an English mathematical reconstruction of a Zhihu question and my answer. Part of the translation work is done by a generative AI tool (ChatGPT 5.6 Sol).

Contents

1 The Morse-theoretic question

1.1 Continuation maps

Let M be a closed manifold. Recall that a *Morse–Smale pair* is a pair (f, g), where $f\colon M \to \mathbb{R}$ is a Morse function and g is a Riemannian metric such that for any two critical points, their stable and unstable manifolds intersect transversely. We work over $\mathbb{F}_2$, so that orientation signs do not appear.
For a Morse–Smale pair (f_i, g_i), write

$$C_i := \mathrm{MC}_*(f_i, g_i), \qquad \partial_i := \partial_{f_i, g_i}.$$

Given two such pairs (f_0, g_0) and (f_1, g_1), choose a regular continuation datum

$$\mathcal{H}^{01} = (f_s, g_s)_{s\in\mathbb{R}}$$

which is constant and equal to (f_0, g_0) for $s \ll 0$, and constant and equal to (f_1, g_1) for $s \gg 0$. Counting isolated solutions of

$$\frac{\mathrm{d}u}{\mathrm{d}s} + \nabla^{g_s} f_s(u(s)) = 0$$

defines a degree-zero map

$$\Phi^{01}\colon C_0 \longrightarrow C_1.$$

The compactification of every one-dimensional continuation moduli space has boundary consisting of a broken Morse trajectory at one end and a continuation trajectory at the other. Algebraically this is

$$\partial_1\Phi^{01} + \Phi^{01}\partial_0 = 0, \tag{1.1}$$

so Φ^{01} is a chain map.
There is a useful way to package Equation (1.1). On the shifted direct sum $C_0[1] \oplus C_1$, consider

$$D_{\mathcal{H}^{01}} = \begin{pmatrix} -\partial_0 & 0 \\ \Phi^{01} & \partial_1 \end{pmatrix}. \tag{1.2}$$

Then

$$D^2_{\mathcal{H}^{01}} = \begin{pmatrix} 0 & 0 \\ -\Phi^{01}\partial_0 + \partial_1\Phi^{01} & 0 \end{pmatrix}.$$

Up to the harmless sign determined by the homological shift convention, the single equation $D^2_{\mathcal{H}^{01}} = 0$ is exactly the chain-map equation.

1.2 A homotopy between continuation maps

Now take three Morse–Smale pairs

$$(f_0, g_0), \qquad (f_1, g_1), \qquad (f_2, g_2)$$

and regular continuation data

$$\mathcal{H}^{01}, \qquad \mathcal{H}^{12}, \qquad \mathcal{H}^{02}.$$

A regular two-parameter family interpolating between the concatenation $\mathcal{H}^{12}\#\mathcal{H}^{01}$ and $\mathcal{H}^{02}$ produces a degree-one map in homological grading,

$$S\colon C_0 \longrightarrow C_2[1],$$

whose parametrized moduli spaces give

$$\partial_2 S + S\partial_0 = \Phi^{12}\Phi^{01} - \Phi^{02}. \tag{1.3}$$

Thus S is a chain homotopy

$$\Phi^{12}\Phi^{01} \simeq \Phi^{02}.$$

Consequently,

$$(\Phi^{12})_*(\Phi^{01})_* = (\Phi^{02})_*$$

on Morse homology. Applying this to a continuation datum followed by a reverse continuation datum, and comparing their concatenation with the constant datum, proves that continuation induces an isomorphism

$$\mathrm{MH}_*(f_0, g_0) \cong \mathrm{MH}_*(f_1, g_1).$$

The same information may again be organized as the square of a block differential. Schematically, one obtains a total complex whose differential contains ∂_i on the diagonal, the continuation maps in the first off-diagonal blocks, and S in a second off-diagonal block. The relevant component of $D^2 = 0$ is precisely Equation (1.3).

1.3 The question

The point is not merely that block-matrix multiplication verifies two familiar identities. Rather, one wants to understand the common structure behind them:

> Why does $D^2 = 0$, resolved into matrix components, first assert that continuation maps are chain maps and then produce chain homotopies between their composites? Is there a single construction that organizes these relations? If one continues to add higher off-diagonal terms, what higher information do the remaining components of $D^2 = 0$ encode?

2 The dg-category of chain complexes

Notation & Convention. From this point onward we use *cohomological grading*. Thus a complex X has a differential $d_X \colon X^n \to X^{n+1}$.

Let $\mathcal{A}$ be an additive category and $X, Y \in \mathsf{Ch}(\mathcal{A})$. Define the graded abelian group

$$\underline{\mathrm{Hom}}^r(X, Y) := \prod_{n \in \mathbb{Z}} \mathrm{Hom}_{\mathcal{A}}(X^n, Y^{n+r}). \tag{2.1}$$

For a homogeneous element $T \in \underline{\mathrm{Hom}}^r(X, Y)$, define

$$\delta T := d_Y T - (-1)^r T d_X. \tag{2.2}$$

Proposition 2.1. *The operator δ satisfies $\delta^2 = 0$. If*

$$T \in \underline{\mathrm{Hom}}^r(X, Y), \qquad U \in \underline{\mathrm{Hom}}^s(Y, Z),$$

then

$$\delta(U \circ T) = \delta U \circ T + (-1)^s U \circ \delta T. \tag{2.3}$$

Proof. Using $d_X^2 = d_Y^2 = 0$, one computes

$$\delta^2 T = d_Y(d_Y T - (-1)^r T d_X) - (-1)^{r+1}(d_Y T - (-1)^r T d_X) d_X = 0.$$

Expanding both sides of Equation (2.3) proves the Leibniz identity. □

The category of complexes over $\mathcal{A}$, enriched in these Hom complexes, is the dg-category $\mathsf{Ch}_{\mathrm{dg}}(\mathcal{A})$ (For the definitions of dg-categories and the dg-nerve, see [6, §2.5].). Its enrichment packages ordinary chain-level notions:

1. A degree-zero element $f \in \underline{\mathrm{Hom}}^0(X, Y)$ is closed if and only if
$$d_Y f - f d_X = 0,$$
that is, if and only if f is a chain map.
2. If $f, g\colon X \to Y$ are chain maps and $h \in \underline{\mathrm{Hom}}^{-1}(X, Y)$, then
$$\delta h = d_Y h + h d_X.$$
Thus $f - g = \delta h$ if and only if h is a chain homotopy from g to f.
3. Elements in degrees $-2, -3, \ldots$ provide the natural receptacles for homotopies between homotopies and their higher analogues.

In particular,

$$H^0\underline{\mathrm{Hom}}^*(X, Y) \cong \mathrm{Hom}_{K(\mathcal{A})}(X, Y), \tag{2.4}$$

where $K(\mathcal{A})$ is the homotopy category of complexes. Equivalently,

$$H^0\mathsf{Ch}_{\mathrm{dg}}(\mathcal{A}) \cong K(\mathcal{A}).$$

For the shift convention $X[m]^n = X^{m+n}$, with differential $d_{X[m]} = (-1)^m d_X$, there are natural identifications

$$\underline{\mathrm{Hom}}^r(X, Y) \cong \underline{\mathrm{Hom}}^0(X, Y[-r]), \qquad H^r\underline{\mathrm{Hom}}^*(X, Y) \cong \mathrm{Hom}_{K(\mathcal{A})}(X, Y[-r]).$$

2.1 The homotopy-theoretic meaning

The nonpositive truncation $\tau_{\leq 0}\underline{\mathrm{Hom}}^*(X, Y)$ given by

$$\cdots \longrightarrow \underline{\mathrm{Hom}}^{-2}(X, Y) \longrightarrow \underline{\mathrm{Hom}}^{-1}(X, Y) \longrightarrow Z^0\underline{\mathrm{Hom}}^*(X, Y).$$

This is a nonpositive cochain complex of abelian groups. After reindexing it as a nonnegative chain complex, the Dold–Kan correspondence $N : \mathsf{sAb} \rightleftarrows \mathsf{Ch}_{\leq 0}(\mathsf{Ab}) : \Gamma$ produces a simplicial abelian group

$$\mathrm{Map}(X, Y) := \Gamma(\tau_{\leq 0}\underline{\mathrm{Hom}}^*(X, Y)) \tag{2.5}$$

satisfying

$$\pi_n\mathrm{Map}(X, Y) \cong H^{-n}\underline{\mathrm{Hom}}^*(X, Y), \qquad n \geq 0. \tag{2.6}$$

These mapping spaces are used to construct the dg-nerve of $\mathsf{Ch}_{\mathrm{dg}}(\mathcal{A})$. When the dg-category is pretriangulated, its dg-nerve is a stable ∞-category, and its homotopy category is $K(\mathcal{A})$. Thus the negative degrees of the Hom complex are not auxiliary bookkeeping: they are the higher homotopy groups of the derived mapping space.

3 Mapping cones and homotopy-commutative squares

3.1 The 2×2 case

Let $f \in \underline{\mathrm{Hom}}^0(X, Y)$, and set $E = X[1] \oplus Y$. On this shifted sum, f becomes a degree-one endomorphism

$$q_f = \begin{pmatrix} 0 & 0 \\ f & 0 \end{pmatrix} \in \underline{\mathrm{End}}^1(E).$$

Deform the direct-sum differential by setting

$$D_E = d_{X[1]} \oplus d_Y + q_f = \begin{pmatrix} -d_X & 0 \\ f & d_Y \end{pmatrix}. \tag{3.1}$$

Then

$$D_E^2 = \begin{pmatrix} 0 & 0 \\ \delta f & 0 \end{pmatrix}.$$

Hence $D_E^2 = 0$ if and only if f is a chain map. In this case (E, D_E) is the mapping cone of f, up to the chosen shift convention. This is exactly the algebraic form of the first continuation-map computation.

3.2 The 4×4 case

Consider a square of degree-zero morphisms

$$\begin{tikzcd} A \arrow[r, "u"] \arrow[d, "v"'] & B \arrow[d, "w"] \\ C \arrow[r, "z"'] & D. \end{tikzcd}$$

Let $S\colon A \to D$ have degree -1. Form

$$E_\square = A[2] \oplus B[1] \oplus C[1] \oplus D$$

and define

$$D_\square = \begin{pmatrix} d_A & 0 & 0 & 0 \\ -u & -d_B & 0 & 0 \\ v & 0 & -d_C & 0 \\ S & w & z & d_D \end{pmatrix}. \tag{3.2}$$

Every entry in Equation (3.2) has total degree one. Direct block multiplication gives

$$D_\square^2 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ \delta u & 0 & 0 & 0 \\ -\delta v & 0 & 0 & 0 \\ \delta S - wu + zv & \delta w & \delta z & 0 \end{pmatrix}. \tag{3.3}$$

Therefore $D_\square^2 = 0$ is equivalent to

$$\delta u = \delta v = \delta w = \delta z = 0, \qquad \delta S = wu - zv. \tag{3.4}$$

The four edges are chain maps, and S is a chain homotopy between the two composites. The Morse relation

$$\Phi^{12}\Phi^{01} \simeq \Phi^{02}$$

is obtained from this pattern by taking one edge of the square to be an identity map.

4 Twisted complexes and the Maurer–Cartan equation

Let $\mathcal{C}$ be a cohomologically graded dg-category and choose finitely many shifted objects

$$E = \bigoplus_i X_i[s_i], \qquad d_E = \bigoplus_i (-1)^{s_i} d_{X_i}.$$

The endomorphism object

$$A_E := \underline{\mathrm{End}}^*(E)$$

is a dg-algebra with differential

$$\delta_E(a) = d_E a - (-1)^{|a|} a d_E. \tag{4.1}$$

Choose a strictly triangular degree-one endomorphism

$$q = (q_{\beta\alpha}) \in \mathfrak{n}_E \subseteq A_E,$$

where $\mathfrak{n}_E$ is the nilpotent ideal of strictly triangular endomorphisms, and put

$$D_E = d_E + q.$$

Because $d_E^2 = 0$ and $|q| = 1$,

$$D_E^2 = d_E q + q d_E + q^2 = \delta_E q + q^2.$$

Thus D_E is a differential precisely when

$$\delta_E q + q^2 = 0. \tag{4.2}$$

This is the Maurer–Cartan equation in the associative dg-algebra A_E. A pair (E, q) satisfying Equation (4.2) is a *twisted complex*. In matrix components, the equation becomes

$$\delta_E q_{\beta\alpha} + \sum_\gamma q_{\beta\gamma} q_{\gamma\alpha} = 0. \tag{4.3}$$

The two terms have different but complementary roles:

- $\delta_E q$ tests whether each individual block is closed;
- q^2 sums the composites along all length-two paths between blocks.

Equation Equation (4.3) requires the differential of a higher block to cancel the appropriate sum of lower composites. The 2×2 mapping cone and the 4×4 homotopy-commutative square are the first two nontrivial instances.

5 Filtration and the hierarchy of equations

Suppose the ordering of the summands of E determines a filtration. Decompose a total differential as

$$D = D_0 + q_1 + q_2 + q_3 + \cdots,$$

where D_0 is the internal differential and q_r lowers filtration by r. Each q_r has total degree one; before the shifts are incorporated, it has degree $1 - r$. Resolving $D^2 = 0$ by filtration degree yields

$$\delta q_r + \sum_{\substack{i+j=r \\ i,j>0}} q_i q_j = 0. \tag{5.1}$$

The first four layers are

$$r=1: \quad \delta q_1 = 0, \tag{5.2}$$
$$r=2: \quad \delta q_2 + q_1^2 = 0, \tag{5.3}$$
$$r=3: \quad \delta q_3 + q_1 q_2 + q_2 q_1 = 0, \tag{5.4}$$
$$r=4: \quad \delta q_4 + q_1 q_3 + q_2^2 + q_3 q_1 = 0. \tag{5.5}$$

Their meanings are as follows:

1. q_1 consists of chain maps.
2. q_2 supplies homotopies among composites of the q_1's.
3. q_3 supplies homotopies among the relations involving q_2.
4. Higher q_r's impose all subsequent coherent compatibilities.

This answers the original question. The equation $D^2 = 0$ is not producing unrelated coincidences. It is a Maurer–Cartan equation in a filtered matrix dg-algebra. Matrix position records the source and target of each operation, while filtration degree records its homotopical order. Resolving the single equation by these two gradings yields chain maps, chain homotopies, homotopies between homotopies, and the entire tower of higher coherence relations.

Remark. The Morse-theoretic part was written homologically, whereas the dg-categorical part was written cohomologically. Over $\mathbb{F}_2$, the distinction causes no sign ambiguity in the displayed Morse identities. Over a general coefficient ring, one must fix orientations and apply the shift and Koszul sign conventions consistently; the Maurer–Cartan description itself is unchanged.

6 My further questions on homotopy and Morse theory

Although the problem itself is purely homological algebra, we would still expect the picture of Morse theory to provide a nice geometrical picture of the Maurer–Cartan equation. This leads to my first question:

> **Question 1:** Is there any rigorous correspondence between the terms of filtration degree decomposition in the Maurer–Cartan equation and the (compactified) moduli space of gradient flow lines? Maurer–Cartan equation and dg-algebras often appear in deformation theory; is there a natural deformation-theoretic illustration of the construction above?

Furthermore, recall that the Morse complex is given by

$$\mathrm{MC}_k(f) := \bigoplus_{p\in\mathrm{Crit}_k(f)} \mathbb{Z}\langle p\rangle, \qquad \partial p = \sum_{q\in\mathrm{Crit}_{k-1}(f)} \#\mathcal{M}(p,q;f)q$$

which only provides us with the signed count of points in the zero-dimensional moduli spaces $\mathcal{M}(p,q;f)$ (note that here $\mathrm{ind}_p(f) - \mathrm{ind}_q(f) = 1$, so the moduli space is indeed zero-dimensional). Even if we know the Morse chain complex completely, we cannot know the information about:

- Handle attachment map;
- The topology of higher-dimensional moduli spaces;
- As we observed in the previous passage, there are higher homotopies in the chain complexes, but we cannot see this from the Morse complex;
- Stable homotopy type of M.

The topic that I am specifically interested in is the relation between stable homotopy.

6.1 Morse theory and stable homotopy

The homological picture of Morse theory is better treated as the "linearized version" of Morse theory, since the most classical version of Morse theory starts from a filtered space. To build the stable homotopy type of a finite CW complex, we need at least:

- Cells in each dimension;
- Stable attaching map of each cell;
- Compatibility between attaching maps.

We may first argue that Morse theory contains information about the stable homotopy type of the manifold; to do this, we may try to find the stable attaching map from the data provided by a Morse function. Note that after rearranging the Morse handles by index, let F_kM denote the subspace obtained by attaching all handles of Morse index at most k. Thus

$$\varnothing = F_{-1}M \subset F_0M \subset \cdots \subset F_nM = M,$$

where F_kM contains all handles with index less than or equal to k. For $p \in \mathrm{Crit}(f)$ and $\lambda(p) := \mathrm{ind}_p(f)$, the k-th filtration quotient is

$$\mathrm{gr}_k M := F_kM/F_{k-1}M \simeq \bigvee_{\lambda(p)=k} S^k.$$

More specifically, for any critical point with index k, the fundamental theorem of Morse theory (handle attachment) gives[1]

$$F_kM \simeq F_{k-1}M \cup_{\alpha_p} \coprod_{f(p)=c} D^k,$$

where $\alpha_k : \coprod_{\lambda(p)=k} S^{k-1} \to F_{k-1}M$ is the handle attaching map and the pushout square is just

$$\begin{array}{ccc}
\coprod_{\lambda(p)=k} S^{k-1} & \xrightarrow{\alpha_k} & F_{k-1}M \\
\downarrow & & \downarrow \\
\coprod_{\lambda(p)=k} D^{k-1} & \longrightarrow & F_kM
\end{array}$$

This leads to the cofiber sequence

$$(*) \qquad \coprod_{\lambda(p)=k} S^{k-1} \xrightarrow{\alpha_k} (F_{k-1}M)_+ \longrightarrow (F_kM)_+ \longrightarrow \bigvee_{\lambda(p)=k} S^k$$

where for a space X, $X_+ := X \sqcup \{*\}$ is the space with a based point added to it. To see further properties of stable homotopy, we may consider the suspension spectrum defined by

$$\Sigma^\infty_+ M = (M_+, \Sigma^1 M_+, \Sigma^2 M_+, \dots)$$

with the structural map $\Sigma(\Sigma^n M_+) \xrightarrow{\cong} \Sigma^\infty M_+$ and $M_+ = M \sqcup \{*\}$. On the homotopy class of maps, we can also write the stabilization

$$[\Sigma^\infty_+ X, \Sigma^\infty_+ Y] := \operatorname*{colim}_{n\geq 0} [\Sigma^n_+ X, \Sigma^n_+ Y]$$

with the transition map be just the suspension $\Sigma : [\Sigma^n_+ X, \Sigma^n_+ Y] \to [\Sigma^{n+1}_+ X, \Sigma^{n+1}_+ Y]$. Setting the spectra

$$\mathbb{S} := \Sigma^\infty S^0, \qquad E_k := \Sigma^\infty_+(F_kM),$$

[1] We can collapse the $n - \lambda(p)$ disk in the handle to attach a $D^{\lambda(p)}$.

where E_k gives a natural filtered spectrum that $\operatorname{colim}_k E_k \simeq E_n = \Sigma^\infty_+ M$. Since Σ^∞_+ preserves homotopy cofibers, one can safely write

$$\begin{aligned}\mathrm{gr}_k(E) &= \mathrm{cofib}(E_{k-1} \to E_k)\\ &\simeq \Sigma^\infty(F_k M/F_{k-1}M) \simeq \bigvee_{\lambda(p)=k} \Sigma^k \mathbb{S}\end{aligned}$$

Since one has the adjunction $\Sigma^\infty_+ :\rightleftarrows: \Omega$, and the left adjoint preserves colimits, one can obtain from stabilizing the cofiber sequence ($*$) that[2]

$$\coprod_{\lambda(p)=k} \Sigma^k \mathbb{S} \xrightarrow{\alpha_k^{\mathrm{st}}} E_{k-1} \longrightarrow E_k \longrightarrow \bigvee_{\lambda(p)=k} \Sigma^k \mathbb{S}$$

Here, we have obtained the stable attaching map from the filtration. This stable homotopic viewpoint leads to the following questions:

> **Question 2:** How do compactified, coherently framed Morse trajectory spaces assemble, via the Pontryagin–Thom construction, into the stable attaching maps—and hence recover $\Sigma^\infty_+ M$?

This is precisely the realization problem underlying the Cohen–Jones–Segal construction [3]; a rigorous construction of the compatible stable normal framings and a proof that the resulting spectrum is $\Sigma^\infty_+ M$ are given in [2, Section 5, Theorem 5.9].

We would construct the stable attaching map using the Pontryagin-Thom construction, as shown in [3].

6.2 Pontryagin-Thom construction and the stable attaching map

From the quotient of the Morse filtration, one can deduce the following based cofiber sequence:

$$E_{k-1} \longrightarrow E_k \longrightarrow \bigvee_{\lambda(p)=k} \Sigma^k \mathbb{S} \xrightarrow{\delta_k} \Sigma E_{k-1}$$

which the rotation in Sp gives $\alpha_k^{\mathrm{st}} \simeq \Sigma^{-1}\delta_k$ and we have the fact that $E_k \simeq \mathrm{cofib}(\alpha_k^{\mathrm{st}})$. Our goal is to construct this stable attaching map from the information provided by Morse theory. It is natural to think about the Pontryagin-Thom construction in this problem.

Recall that the Pontryagin-Thom construction states that, via the construction of the Thom space of the trivial bundle $\mathrm{Th}(\mathbb{R}^N) \cong B_+ \wedge S^N$ over the base space B. With a embedding of a closed manifold $i : M^d \to \mathbb{R}^{d+N}$ and its normal bundle ν_i, the corresponding tubular neighborhood embedding $\tau : D(\nu_i) \to \mathbb{R}^{d+N}$ leads to the Thom collapse map $c_i : S^{d+N} \to \mathrm{Th}(\nu_i)$ defined by

$$c_i(x) := \begin{cases} [\tau^{-1}(x)], & x \in \tau\big(D(\nu_i)\big),\\ *, & x \notin \tau\big(D(\nu_i)\big).\end{cases}$$

If we further consider the framing $\varphi : \nu_i \to M \times \mathbb{R}^N$ on the submanifold, then one can consider $\mathrm{Th}(\nu_i) \cong M_+ \wedge S^N$ and the augmentation $\epsilon_M : M_+ \to S^0$, which sends the base point to the base point and other points to the non-base point; we can get

$$M_+ \wedge S^N \xrightarrow{\epsilon_M \wedge \mathrm{id}} S^0 \wedge S^N \cong S^N.$$

[2] Another way to see this is to consider the functor $\Sigma^\infty : \mathcal{S}_* \to \mathsf{Sp}$, then since Sp is a stable ∞-category, rotating the stable quotient cofiber sequence gives the same result.

We may define the framed Pontryagin-Thom map:

$$\begin{aligned}\mathrm{PT}(M,i) : S^{d+N} &\xrightarrow{c_i} \mathrm{Th}(\nu_i)\\ &\xrightarrow{\mathrm{Th}(\varphi)} M_+ \wedge S^N\\ &\xrightarrow{\epsilon_M \wedge \mathrm{id}} S^0 \wedge S^N \cong S^N.\end{aligned}$$

And Thom's work has proved that this induces (after stabilization) the isomorphism between the cobordism group and stable homotopy groups:

$$\Omega_d^{\mathrm{fr}} \cong \pi_d^S.$$

One needs to point out that this construction is stable since $\mathrm{PT}(M, i\times 0, \varphi\oplus 1) \simeq \Sigma\mathrm{PT}(M,i,\varphi)$; the proof is simply

$$S^{d+N+1} \cong S^{d+N} \wedge S^1 \xrightarrow{\mathrm{PT}(M)\wedge\mathrm{id}} S^N \wedge S^1 \cong S^{N+1}.$$

One may also check that the Pontryagin-Thom construction is well-defined (with respect to bordism and homotopy) and compatible with the group structure.

Then we can turn our eyes to Morse theory. To collect more geometrical information given in the gradient flow and Morse moduli spaces, people have defined the Morse flow category [2, 4] [5, Section 3]:

Definition 6.1 (Morse Flow Category)**.** Defined the Top-enriched category $\mathcal{C}_f$, with the object defined by

$$\mathrm{ob}(\mathcal{C}_f) = \mathrm{Crit}(f)$$

and the morphism defined by

$$\mathrm{mor}(\mathcal{C}_f) = \begin{cases} \overline{\mathcal{M}}(p,q), & p\neq q,\ f(p) > f(q),\\ \{\mathrm{id}_p\}, & p = q,\\ \varnothing, & \text{otherwise.}\end{cases}$$

For $f(p) > f(r) > f(q)$, the composition is boundary-face inclusion

$$\begin{aligned}\overline{\mathcal{M}}(p,r)\times\overline{\mathcal{M}}(r,q) &\longrightarrow \overline{\mathcal{M}}(p,q)\\ (\gamma_1,\gamma_2) &\longmapsto (\gamma_1,\gamma_2)\end{aligned}$$

Note that the composition demands the argument of the broken flow line, and more generally, one can define the general flow category

Definition 6.2 (Flow Category)**.** The flow category is defined by the Top-enriched category $\mathcal{C}_f$, with the objects defined by a finite point set with grading

$$\mathrm{ob}(\mathcal{C}_f) = \mathrm{Crit}(f), \qquad |\cdot| : \mathrm{ob}(\mathcal{C}_F) \to \mathbb{Z}$$

and the morphism defined by

$$\mathrm{mor}(\mathcal{C}_f) = \begin{cases} \overline{\mathcal{M}}(p,q), & p\neq q,\ |p| > |q|,\\ \{\mathrm{id}_p\}, & p = q,\\ \varnothing, & \text{otherwise,}\end{cases}$$

where $\overline{\mathcal{M}}(p,q)$ is a compact manifold with corners with $\dim\overline{\mathcal{M}}(p,q) := |p|-|q|-1$. For $f(p) > f(r) > f(q)$, the composition is boundary-face inclusion

$$\begin{aligned}\overline{\mathcal{M}}(p,r)\times\overline{\mathcal{M}}(r,q) &\longrightarrow \overline{\mathcal{M}}(p,q)\\ (\gamma_1,\gamma_2) &\longmapsto (\gamma_1,\gamma_2)\end{aligned}$$

and we need the corner identification compatible with composition.

The boundary-face condition in the preceding definition should be understood as follows. For every intermediate object r, let

$$F_r(p,q) := \operatorname{im}\left(\overline{\mathcal{M}}(p,r) \times \overline{\mathcal{M}}(r,q) \longrightarrow \overline{\mathcal{M}}(p,q)\right).$$

Then

$$F_r(p,q) \cong \overline{\mathcal{M}}(p,r) \times \overline{\mathcal{M}}(r,q)$$

is a closed boundary face and

$$\partial\overline{\mathcal{M}}(p,q) = \bigcup_{p \succ r \succ q} F_r(p,q).$$

This is generally not a disjoint union, since distinct boundary faces intersect along higher-codimensional corners. More precisely, a chain

$$p = p_0 \succ p_1 \succ \cdots \succ p_\ell = q$$

determines a codimension-$(\ell-1)$ corner

$$\prod_{j=0}^{\ell-1} \overline{\mathcal{M}}(p_j, p_{j+1}) \subset \overline{\mathcal{M}}(p,q).$$

Compatibility of the corner identifications with composition means that the inclusion of this corner is independent of how the corresponding iterated composition is parenthesized.

To apply the Pontryagin–Thom construction, the structure of a flow category alone is not sufficient. We must equip all morphism spaces with mutually compatible neat embeddings and stable normal framings.

Definition 6.3 (Framed flow category)**.** A framed flow category is a flow category $\mathcal{C}$ together with the following data.

For every pair $p \succ q$, choose a Euclidean orthant

$$\mathbb{E}_{p,q} = \mathbb{R}^{a_{p,q}} \times [0,\infty)^{c_{p,q}}$$

and a neat embedding

$$\iota_{p,q} : \overline{\mathcal{M}}(p,q) \hookrightarrow \mathbb{E}_{p,q}.$$

The ambient orthants are equipped with compatible face decompositions. In a neighborhood of the face indexed by an intermediate object r, we require an identification

$$\mathbb{E}_{p,q} \cong \mathbb{E}_{p,r} \times [0,\infty)_r \times \mathbb{E}_{r,q},$$

under which

$$\partial^r \mathbb{E}_{p,q} = \mathbb{E}_{p,r} \times \{0\} \times \mathbb{E}_{r,q}.$$

The embedding is required to satisfy

$$\iota_{p,q}^{-1}(\partial^r \mathbb{E}_{p,q}) = F_r(p,q)$$

and, under the identification

$$F_r(p,q) \cong \overline{\mathcal{M}}(p,r) \times \overline{\mathcal{M}}(r,q),$$

its restriction is

$$\iota_{p,q}|_{F_r(p,q)} = \iota_{p,r} \times \{0\} \times \iota_{r,q}.$$

Moreover, every corner stratum is required to meet the corresponding ambient stratum perpendicularly.

Let $\nu_{p,q} \longrightarrow \overline{\mathcal{M}}(p,q)$ be the normal bundle of $\iota_{p,q}$. After a sufficiently large coherent stabilization of the ambient spaces, choose a vector space $V_{p,q}$ and a normal framing

$$\varphi_{p,q} : \nu_{p,q} \xrightarrow{\cong} \underline{V}_{p,q}.$$

For every $p \succ r \succ q$, the normal bundles satisfy

$$\nu_{p,q}|_{F_r(p,q)} \cong \mathrm{pr}_1^* \nu_{p,r} \oplus \mathrm{pr}_2^* \nu_{r,q}.$$

The vector spaces and framings are required to satisfy $V_{p,q} \cong V_{p,r} \oplus V_{r,q}$ and

$$\varphi_{p,q}|_{F_r(p,q)} = \mathrm{pr}_1^* \varphi_{p,r} \oplus \mathrm{pr}_2^* \varphi_{r,q}.$$

These identifications must themselves be compatible on all higher corners.

The absence of an additional trivial line in the normal-bundle decomposition is important. Indeed, along $F_r(p,q)$, the collar direction gives

$$T\overline{\mathcal{M}}(p,q)\Big|_{F_r(p,q)} \cong T\overline{\mathcal{M}}(p,r) \oplus T\overline{\mathcal{M}}(r,q) \oplus \mathbb{R}_\rho,$$

while the ambient tangent bundle satisfies

$$T\mathbb{E}_{p,q}|_{F_r(p,q)} \cong T\mathbb{E}_{p,r} \oplus T\mathbb{E}_{r,q} \oplus \mathbb{R}_\rho.$$

The two collar directions cancel upon passing to the normal quotient.
We now apply the Pontryagin–Thom construction to this coherent system. Here

$$\mathbb{E}_{p,q}^+ = \mathbb{E}_{p,q} \cup \{\infty\}$$

denotes the one-point compactification, whereas a subscript $+$ continues to denote the adjunction of a disjoint base point. Similarly,

$$V_{p,q}^+ = S^{V_{p,q}}.$$

Choose tubular neighborhoods which are compatible with all boundary-face decompositions. The resulting Thom collapse is $c_{p,q} : \mathbb{E}_{p,q}^+ \longrightarrow \mathrm{Th}(\nu_{p,q})$. Using the framing, we obtain

$$\mathrm{Th}(\nu_{p,q}) \cong \overline{\mathcal{M}}(p,q)_+ \wedge V_{p,q}^+.$$

Thus the framed Pontryagin–Thom map is

$$\begin{aligned} \mathrm{PT}_{p,q} : \mathbb{E}_{p,q}^+ &\xrightarrow{c_{p,q}} \mathrm{Th}(\nu_{p,q}) \\ &\xrightarrow{\mathrm{Th}(\varphi_{p,q})} \overline{\mathcal{M}}(p,q)_+ \wedge V_{p,q}^+ \\ &\xrightarrow{\epsilon_{\overline{\mathcal{M}}(p,q)} \wedge \mathrm{id}} V_{p,q}^+. \end{aligned}$$

The coherence of the embeddings, tubular neighborhoods, and framings implies a corresponding compatibility of the Pontryagin–Thom maps. Under the identifications

$$(\partial^r \mathbb{E}_{p,q})^+ \cong \mathbb{E}_{p,r}^+ \wedge \mathbb{E}_{r,q}^+$$

and

$$V_{p,q}^+ \cong V_{p,r}^+ \wedge V_{r,q}^+,$$

we have

$$\mathrm{PT}_{p,q}|_{(\partial^r \mathbb{E}_{p,q})^+} = \left(\mathrm{id}_{V_{p,r}^+} \wedge \mathrm{PT}_{r,q}\right) \circ \left(\mathrm{PT}_{p,r} \wedge \mathrm{id}_{\mathbb{E}_{r,q}^+}\right). \tag{6.1}$$

Indeed, the tubular neighborhood of $F_r(p,q)$ is the product of the tubular neighborhoods of $\overline{\mathcal{M}}(p,r)$ and $\overline{\mathcal{M}}(r,q)$, and the Thom collapse of a product is the smash product of the corresponding Thom collapses.

It is essential that $\mathrm{PT}_{p,q}$ be interpreted as a relative, or cubical, Pontryagin–Thom map. The lower-dimensional Pontryagin–Thom maps determine a boundary map

$$\beta_{p,q} : (\partial \mathbb{E}_{p,q})^+ \longrightarrow V_{p,q}^+,$$

and $\mathrm{PT}_{p,q}$ is an extension satisfying

$$\mathrm{PT}_{p,q}|_{(\partial \mathbb{E}_{p,q})^+} = \beta_{p,q}.$$

In general, $\beta_{p,q}$ is not the constant map. Consequently, $\mathrm{PT}_{p,q}$ does not descend to a map

$$\mathbb{E}_{p,q}^+/(\partial \mathbb{E}_{p,q})^+ \longrightarrow V_{p,q}^+.$$

Moreover, when $\mathbb{E}_{p,q}$ contains an orthant factor, $\mathbb{E}_{p,q}^+$ is not a sphere. For example,

$$\big(\mathbb{R}^a \times [0,\infty)\big)^+ \cong D^{a+1},$$

with the point at infinity lying on the boundary. Hence an individual $\mathrm{PT}_{p,q}$ does not, in general, determine an element of a stable homotopy group.

In the boundary-free case, the construction reduces to the ordinary Pontryagin–Thom construction. If

$$\dim \overline{\mathcal{M}}(p,q) = |p| - |q| - 1 = d$$

and

$$\overline{\mathcal{M}}(p,q) \hookrightarrow \mathbb{R}^{d+N}$$

is a framed embedding, then the collapse gives

$$S^{d+N} \longrightarrow S^N,$$

and hence an element

$$[\overline{\mathcal{M}}(p,q)]_{\mathrm{fr}} \in \pi_d^S.$$

After shifting by $|q|$, this may be regarded as a stable map

$$\Sigma^{|p|-1}\mathbb{S} \longrightarrow \Sigma^{|q|}\mathbb{S}.$$

When the moduli space has boundary, however, the corresponding datum must be retained together with all its prescribed boundary values.

This gives a useful interpretation of the dimensions of the moduli spaces. A zero-dimensional moduli space produces a basic stable map, a one-dimensional compactified moduli space produces a homotopy between composites, and a two-dimensional compactified moduli space produces a homotopy between such homotopies. Higher-dimensional moduli spaces similarly encode higher coherence.

For example, suppose that

$$|p| = k+2, \qquad |q| = k.$$

Then $\overline{\mathcal{M}}(p,q)$ is one-dimensional, and

$$\partial \overline{\mathcal{M}}(p,q) = \bigcup_{|r|=k+1} \mathcal{M}(p,r) \times \mathcal{M}(r,q).$$

At each boundary point, Equation (6.1) identifies the boundary value of $\mathrm{PT}_{p,q}$ with the composite of the two zero-dimensional Pontryagin–Thom maps. Thus the framed one-dimensional moduli space supplies a coherent filling of the total boundary datum. After forgetting the framing and retaining only the orientation, this reduces to

$$\sum_{|r|=k+1} \#\mathcal{M}(p,r)\,\#\mathcal{M}(r,q) = 0,$$

which is precisely the relation

$$\partial^2_{\mathrm{Morse}} = 0.$$

The coherent Pontryagin–Thom construction therefore retains strictly more information than the Morse differential.

We can now assemble the relative Pontryagin–Thom maps into genuine stable attaching maps. Choose a sufficiently large uniform suspension shift D. For each object p, the Cohen–Jones–Segal realization assigns a cell block B_p of dimension

$$\dim B_p = D + |p|.$$

For every $q \prec p$, a framed tubular neighborhood of $\overline{\mathcal{M}}(p,q)$ determines a region in ∂B_p, and the map $\mathrm{PT}_{p,q}$ identifies this region with the corresponding part of the lower cell block B_q. The complement of all such tubular regions is sent to the base point.

Equation (6.1) guarantees that these identifications agree on overlaps. Indeed, on a corner corresponding to

$$p \succ r \succ q,$$

one may either first attach through r and then through q, or use the restriction of the p-to-q Pontryagin–Thom map. The two procedures coincide by Equation (6.1). Compatibility on higher corners shows that all iterated identifications are independent of parenthesization. Therefore the resulting quotient relation is well-defined.

Let $X^{\mathrm{CJS}}_{\le k}$ denote the subcomplex formed by the cells corresponding to objects of grading at most k. Its k-th filtration quotient is

$$X^{\mathrm{CJS}}_{\le k}/X^{\mathrm{CJS}}_{\le k-1} \simeq \bigvee_{|p|=k} S^{D+k}.$$

Define the corresponding filtered spectrum by

$$E^{\mathrm{CJS}}_{\le k} := \Sigma^{-D}\Sigma^{\infty} X^{\mathrm{CJS}}_{\le k}.$$

It follows that

$$E^{\mathrm{CJS}}_{\le k}/E^{\mathrm{CJS}}_{\le k-1} \simeq \bigvee_{|p|=k} \Sigma^k \mathbb{S}_p.$$

The attaching maps of the cell blocks therefore assemble into a stable map

$$\alpha^{\mathrm{PT}}_k : \bigvee_{|p|=k} \Sigma^{k-1}\mathbb{S}_p \longrightarrow E^{\mathrm{CJS}}_{\le k-1},$$

and there is a cofiber sequence

$$\bigvee_{|p|=k} \Sigma^{k-1}\mathbb{S}_p \xrightarrow{\alpha^{\mathrm{PT}}_k} E^{\mathrm{CJS}}_{\le k-1} \longrightarrow E^{\mathrm{CJS}}_{\le k}.$$

Consequently,

$$E^{\mathrm{CJS}}_{\le k} \simeq \mathrm{cofib}(\alpha^{\mathrm{PT}}_k).$$

Thus an individual relative map $\mathrm{PT}_{p,q}$ is not itself the stable attaching map. Rather, the entire coherent system

$$\{\mathrm{PT}_{p,q}\}_{p\succ q}$$

assembles, through the Cohen–Jones–Segal quotient, into the global stable attaching maps α^{PT}_k.

It remains to compare these maps with the stable attaching maps obtained abstractly from the Morse filtration. Let $\overline{\mathcal{W}}^u(p)$ denote the compactification of the unstable manifold of p, equivalently the moduli space of trajectories starting at p with a free terminal point. Its boundary has the decomposition

$$\partial\overline{\mathcal{W}}^u(p) = \bigcup_{p\succ q} \overline{\mathcal{M}}(p,q) \times \overline{\mathcal{W}}^u(q).$$

Evaluation at the free endpoint defines

$$\mathrm{ev}_p : \overline{\mathcal{W}}^u(p) \longrightarrow F_{|p|}M.$$

The preceding boundary formula is compatible with the boundary decomposition of the CJS cell block. Applying the coherent Pontryagin–Thom construction to these compactified unstable manifolds produces a filtered stable equivalence.

Theorem 6.4 (Morse–CJS comparison)**.** *Let (f, g) be a Morse–Smale pair on a closed manifold M, and equip its Morse flow category with coherent normal data compatible with the Morse handle decomposition. Then there are filtered stable equivalences*

$$\Phi_k : E^{\mathrm{CJS}}_{\le k} \xrightarrow{\simeq} \Sigma^\infty_+ F_k M = E_k$$

which identify the filtration quotients

$$\bigvee_{|p|=k} \Sigma^k \mathbb{S}_p \simeq E^{\mathrm{CJS}}_{\le k} / E^{\mathrm{CJS}}_{\le k-1}$$

with

$$\bigvee_{\lambda(p)=k} \Sigma^k \mathbb{S}_p \simeq E_k / E_{k-1}.$$

Under these identifications, the corresponding stable attaching maps agree:

$$\Phi_{k-1} \circ \alpha^{\mathrm{PT}}_k \simeq \alpha^{\mathrm{st}}_k.$$

Consequently,

$$E^{\mathrm{CJS}}_{\le k} \simeq \Sigma^\infty_+ F_k M$$

for every k, and hence

$$E^{\mathrm{CJS}}(\mathcal{C}_f) \simeq \Sigma^\infty_+ M.$$

The equality of the attaching maps follows formally from the naturality of the connecting morphisms in the two filtered cofiber sequences. More precisely, the filtered equivalences Φ_k identify the two quotient maps, and therefore identify their connecting morphisms

$$\delta^{\mathrm{CJS}}_k \qquad \text{and} \qquad \delta_k.$$

After desuspension, this gives

$$\alpha^{\mathrm{PT}}_k = \Sigma^{-1} \delta^{\mathrm{CJS}}_k \simeq \Sigma^{-1} \delta_k = \alpha^{\mathrm{st}}_k,$$

up to the standard sign convention for rotating cofiber sequences.

6.3 $H\mathbb{Z}$-linearization and the geometric Maurer–Cartan equation

From this point onward, we assume that the Morse moduli spaces have been coherently oriented, so that their fundamental chains and signed counts are defined over $\mathbb{Z}$. Without coherent orientations, one should replace $H\mathbb{Z}$ throughout by $H\mathbb{F}_2$.

The unit map of ring spectra

$$\eta : \mathbb{S} \longrightarrow H\mathbb{Z}$$

induces the extension-of-scalars functor

$$H\mathbb{Z} \wedge - : \mathsf{Sp} \longrightarrow \mathrm{Mod}_{H\mathbb{Z}}.$$

The stable ∞-category of $H\mathbb{Z}$-modules is equivalent to the derived ∞-category of chain complexes of abelian groups:

$$\mathrm{Mod}_{H\mathbb{Z}} \simeq D(\mathbb{Z}).$$

Under this equivalence, $H\mathbb{Z} \wedge \Sigma^\infty_+ X$ corresponds to the singular chain complex $C_*(X;\mathbb{Z})$. Thus smashing with $H\mathbb{Z}$ may be regarded as the passage from stable homotopy data to its derived $H\mathbb{Z}$-linear approximation. Apply this functor to the filtered CJS spectrum and define

$$\mathsf{C}^{H\mathbb{Z}}_{\le k} := H\mathbb{Z} \wedge E^{\mathrm{CJS}}_{\le k}.$$

Since smashing with $H\mathbb{Z}$ preserves cofiber sequences, we have

$$\mathsf{C}^{H\mathbb{Z}}_{\le k}/\mathsf{C}^{H\mathbb{Z}}_{\le k-1} \simeq \bigvee_{|p|=k} \Sigma^k H\mathbb{Z}_p.$$

Consequently,

$$\mathrm{gr}_k\, \mathsf{C}^{H\mathbb{Z}} \simeq \bigvee_{|p|=k} \Sigma^k H\mathbb{Z}_p.$$

The Morse–CJS comparison gives a filtered equivalence

$$H\mathbb{Z} \wedge E^{\mathrm{CJS}}_{\le \bullet} \simeq H\mathbb{Z} \wedge \Sigma^\infty_+ F_\bullet M.$$

Under the equivalence $\mathrm{Mod}_{H\mathbb{Z}} \simeq D(\mathbb{Z})$, the right-hand side is represented by the cellular chain complex of the Morse filtration. Hence there are filtered quasi-isomorphisms

$$C^{\mathrm{Morse}}_*(f;\mathbb{Z}) \simeq C^{\mathrm{cell}}_*(F_\bullet M;\mathbb{Z}) \simeq C^{\mathrm{sing}}_*(M;\mathbb{Z}).$$

In this sense, the Morse complex is the $H\mathbb{Z}$-linearization of the filtered CJS spectrum.

The loss of information under this linearization can be seen directly from the stable attaching maps. Suppose

$$|p| - |q| = r.$$

A component of the stable attaching map from the p-cell to the q-cell is represented by a stable map

$$\Sigma^{|p|-1}\mathbb{S} \longrightarrow \Sigma^{|q|}\mathbb{S}$$

and hence by an element of π^S_{r-1}. After smashing with $H\mathbb{Z}$, it becomes a map

$$\Sigma^{|p|-1}H\mathbb{Z} \longrightarrow \Sigma^{|q|}H\mathbb{Z},$$

whose homotopy class belongs to $\pi_{r-1}(H\mathbb{Z})$. Since

$$\pi_j(H\mathbb{Z}) = \begin{cases} \mathbb{Z}, & j = 0, \\ 0, & j \neq 0, \end{cases}$$

only the case $r = 1$ survives. In this case, the framed Pontryagin–Thom map of the zero-dimensional moduli space is sent to its signed count

$$\#\mathcal{M}(p,q) \in \mathbb{Z}.$$

For $r > 1$, the map

$$\pi^S_{r-1} \longrightarrow \pi_{r-1}(H\mathbb{Z})$$

is zero. Thus the $H\mathbb{Z}$-linearization retains the Morse differential but forgets the higher stable attaching information.

There is nevertheless a chain-level construction which retains the full geometry of the compactified trajectory spaces and exhibits their boundary relations as a Maurer–Cartan equation. For each $p \succ q$, consider

$$C_{-*}\big(\overline{\mathcal{M}}(p,q);\mathbb{Z}\big),$$

where a chain of homological degree j is placed in cohomological degree $-j$. If

$$\sigma \in C_j\big(\overline{\mathcal{M}}(p,q);\mathbb{Z}\big),$$

define its total degree by

$$|\sigma|_{\mathrm{tot}} := |p| - |q| - j.$$

The cross product followed by the boundary-face inclusion defines composition:

$$\begin{aligned} C_{-*}\big(\overline{\mathcal{M}}(p,r)\big) \otimes C_{-*}\big(\overline{\mathcal{M}}(r,q)\big) &\xrightarrow{\times} C_{-*}\big(\overline{\mathcal{M}}(p,r) \times \overline{\mathcal{M}}(r,q)\big) \\ &\longrightarrow C_{-*}\big(\overline{\mathcal{M}}(p,q)\big). \end{aligned}$$

One may use cubical chains, or another strictly associative monoidal chain model, so that this composition is strictly associative.

Choose coherent fundamental chains

$$\mu_{p,q} \in C_{|p|-|q|-1}\big(\overline{\mathcal{M}}(p,q);\mathbb{Z}\big)$$

compatible with the product decompositions of all boundary faces. Their total degrees are

$$|\mu_{p,q}|_{\mathrm{tot}} = |p| - |q| - \big(|p| - |q| - 1\big) = 1.$$

Thus

$$\mu := \sum_{p \succ q} \mu_{p,q}$$

is a degree-one strictly upper-triangular element in the matrix dg-algebra formed from these morphism complexes.

The oriented boundary decomposition

$$\partial\overline{\mathcal{M}}(p,q) = \bigcup_{p \succ r \succ q} \overline{\mathcal{M}}(p,r) \times \overline{\mathcal{M}}(r,q)$$

implies

$$\partial\mu_{p,q} + \sum_{p \succ r \succ q} (-1)^{\epsilon(p,r,q)} \mu_{r,q} \circ \mu_{p,r} = 0.$$

After incorporating the orientation and Koszul signs into the definition of composition, this becomes

$$\partial\mu_{p,q} + \sum_{p \succ r \succ q} \mu_{r,q} \circ \mu_{p,r} = 0.$$

Equivalently, in matrix form,

$$\partial\mu + \mu^2 = 0. \tag{6.2}$$

Thus the compactified trajectory spaces themselves determine a geometric Maurer–Cartan element.

To compare Equation (6.2) with the filtration hierarchy introduced earlier, decompose

$$\mu = \mu_1 + \mu_2 + \mu_3 + \cdots, \qquad \mu_r := \sum_{|p|-|q|=r} \mu_{p,q}.$$

Since

$$|p| - |q| = r \quad \Longleftrightarrow \quad \dim \overline{\mathcal{M}}(p, q) = r - 1,$$

the term μ_r is precisely the contribution of the $(r-1)$-dimensional compactified moduli spaces. Resolving Equation (6.2) by filtration degree gives

$$\partial \mu_r + \sum_{\substack{i+j=r \\ i,j>0}} \mu_i \mu_j = 0. \tag{6.3}$$

The first equations are

$$\begin{aligned} r = 1: &\quad \partial \mu_1 = 0, \\ r = 2: &\quad \partial \mu_2 + \mu_1^2 = 0, \\ r = 3: &\quad \partial \mu_3 + \mu_1 \mu_2 + \mu_2 \mu_1 = 0, \\ r = 4: &\quad \partial \mu_4 + \mu_1 \mu_3 + \mu_2^2 + \mu_3 \mu_1 = 0. \end{aligned}$$

These equations have exactly the same form as the filtration-degree decomposition

$$\delta q_r + \sum_{\substack{i+j=r \\ i,j>0}} q_i q_j = 0$$

of the algebraic Maurer–Cartan equation. The geometric correspondence is therefore

$$\text{filtration drop } r \quad \longleftrightarrow \quad \text{moduli-space dimension } r - 1.$$

The ordinary Morse complex is obtained from this enriched construction by retaining only the zero-dimensional moduli spaces and applying the signed count

$$C_0(\mathcal{M}(p, q); \mathbb{Z}) \longrightarrow \mathbb{Z}.$$

It therefore retains μ_1, which becomes the Morse differential, but discards the positive-dimensional chains $\mu_2, \mu_3, \ldots$. The equation

$$\partial \mu_2 + \mu_1^2 = 0$$

then reduces, after taking the signed count of the boundary of each compact one-dimensional moduli space, to

$$\partial_{\text{Morse}}^2 = 0.$$

This should be distinguished from the nontrivial higher terms $q_2, q_3, \ldots$ which appeared earlier in the totalization of continuation maps. For a single minimal Morse complex, the positive-dimensional trajectory spaces are not represented as additional differential blocks. For a diagram of Morse complexes, however, the mapping complexes

$$\operatorname{Hom}^\bullet(C_i, C_j)$$

contain negative-degree elements. In this setting, q_1 consists of continuation chain maps, q_2 consists of chain homotopies, and the higher q_r's are produced by higher parameterized continuation moduli spaces. Thus the two constructions are manifestations of the same coherence principle, but they are not literally the same collection of maps.

The Maurer–Cartan description also has a natural deformation-theoretic interpretation. Let

$$G = \bigoplus_i X_i[s_i]$$

be a fixed associated graded object, and let

$$\mathfrak{n} \subset \operatorname{End}^\bullet(G)$$

be the nilpotent dg-algebra of strictly filtration-lowering endomorphisms. For $q \in \mathfrak{n}^1$, the operator

$$D = d_G + q$$

is a differential if and only if

$$\delta q + q^2 = 0.$$

Hence Maurer–Cartan elements parameterize filtered differentials on G which induce the original differential on the associated graded object.

The group

$$1 + \mathfrak{n}^0$$

acts on the set of Maurer–Cartan elements by filtered changes of splitting:

$$q \longmapsto g(d_G + q)g^{-1} - d_G.$$

Thus Maurer–Cartan elements modulo gauge equivalence classify filtered twisted complexes with fixed associated graded object.

This description gives a successive obstruction theory. Suppose that

$$q_{<r} = q_1 + \cdots + q_{r-1}$$

has been constructed and satisfies the Maurer–Cartan equation through filtration degree $r-1$. Define

$$\mathcal{O}_r := \sum_{\substack{i+j=r \\ i,j>0}} q_i q_j.$$

The lower Maurer–Cartan equations imply $\delta\mathcal{O}_r = 0$, thus the term $\mathcal{O}_r$ determines an obstruction class $[\mathcal{O}_r] \in H^2\left(\operatorname{gr}_F^r \mathfrak{n}, \delta\right)$. A term q_r satisfying

$$\delta q_r = -\mathcal{O}_r$$

exists if and only if $[\mathcal{O}_r] = 0$. When it exists, the choices of q_r, modulo the corresponding gauge equivalence, form a torsor controlled by $H^1\left(\operatorname{gr}_F^r \mathfrak{n}, \delta\right)$.

Geometrically, the obstruction cycle

$$\mathcal{O}_r = \sum_{i+j=r} \mu_i \mu_j$$

is the sum of all broken boundary faces of the $(r-1)$-dimensional compactified moduli spaces. The fundamental chain μ_r supplies the geometric nullhomotopy

$$\partial \mu_r = -\mathcal{O}_r.$$

Thus the moduli spaces themselves geometrically solve the successive Maurer–Cartan obstruction problems.

At the sphere-spectrum level, one should not expect an ordinary dg-algebra Maurer–Cartan equation without first choosing an algebraic model. The appropriate objects are homotopy-coherent cochain complexes in the stable ∞-category Sp, equivalently suitable filtered spectra. Their successive coherence obstructions are expressed by Toda brackets and higher Toda brackets. After $H\mathbb{Z}$-linearization, these become the corresponding dg-algebraic obstruction classes and, when defined, Massey products. The framed moduli spaces provide geometric representatives and nullbordisms for this spectral coherence data.

6.4 Other Problems

The preceding discussion suggests an inverse problem. For a genuine Morse flow category, the CJS construction already provides a stable lift of the Morse complex. The problem becomes nontrivial when one starts only with an abstract filtered $H\mathbb{Z}$-module or filtered chain complex.

Let

$$\mathrm{Fil}^{\mathrm{fin}}_{\mathrm{cell}}(\mathsf{Sp}^{\omega})$$

denote the ∞-category of finite filtered spectra whose associated graded pieces are finite wedges of suspended sphere spectra. For a finite filtered perfect $H\mathbb{Z}$-module C, define its space of stable enhancements by

$$\mathrm{Lift}_{\mathbb{S}}(C) := \mathrm{hofib}_C \left(H\mathbb{Z} \wedge - : \mathrm{Fil}^{\mathrm{fin}}_{\mathrm{cell}}(\mathsf{Sp}^{\omega}) \longrightarrow \mathrm{Fil}^{\mathrm{fin}}(\mathrm{Perf}(H\mathbb{Z})) \right).$$

Question 3 (Stable enhancement and obstruction): Let C be a finite filtered perfect $H\mathbb{Z}$-module such that

$$\mathrm{gr}_k\, C \simeq \bigvee_{p \in P_k} \Sigma^k H\mathbb{Z}.$$

When is

$$\mathrm{Lift}_{\mathbb{S}}(C) \neq \varnothing,$$

and how can one describe its connected components and higher homotopy type? Can its successive obstruction and ambiguity classes be expressed spectrally in terms of Toda brackets, algebraically in terms of Maurer–Cartan obstruction classes and Massey products, and geometrically in terms of framed compactified moduli spaces and their corner decompositions?

Equivalently, one may ask when an $H\mathbb{Z}$-linear twisted complex

$$(G, q), \qquad \delta q + q^2 = 0,$$

admits an $\mathbb{S}$-linear homotopy-coherent enhancement whose $H\mathbb{Z}$-linearization recovers (G, q). A framed flow category should then be regarded as a geometric presentation of such an enhancement, rather than as merely another name for a filtered chain complex.

There is a logically distinct geometric recognition problem.

Question 4 (Geometric recognition): Which framed flow categories lie in the geometric essential image of the assignment

$$(M, f, g) \longmapsto \mathcal{C}_f$$

from Morse–Smale systems on closed smooth manifolds, before passing to the stable localization of flow categories? Equivalently, what additional geometric conditions distinguish genuine Morse flow categories from arbitrary framed flow categories presenting the same spectrum?

The theorem of Abouzaid–Blumberg that the stable ∞-category of framed flow categories is equivalent to the ∞-category of spectra does not make Question 4 vacuous [1]. Their theorem concerns the category obtained after passing to an appropriate stable bordism localization. It does not assert that every framed flow category is geometrically realized, before localization, by compactified trajectory spaces of a Morse–Smale system on a closed manifold.

After passing completely to stable homotopy types, one obtains the related but weaker recognition problem:

for which finite spectra X does there exist a closed smooth manifold M such that $X \simeq \Sigma^{\infty}_{+} M$?

This problem concerns the essential image of suspension spectra of closed smooth manifolds and is different from the geometric realization of a particular framed flow category. Thus Question 3 asks when algebraic $H\mathbb{Z}$-linear data admit a stable and geometric enhancement, whereas Question 4 asks which such enhancements genuinely arise from Morse geometry.

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